

- 5 (a) Two light sources that produce light with the same wavelength are placed at position A and B respectively as shown in Fig. 5.1.

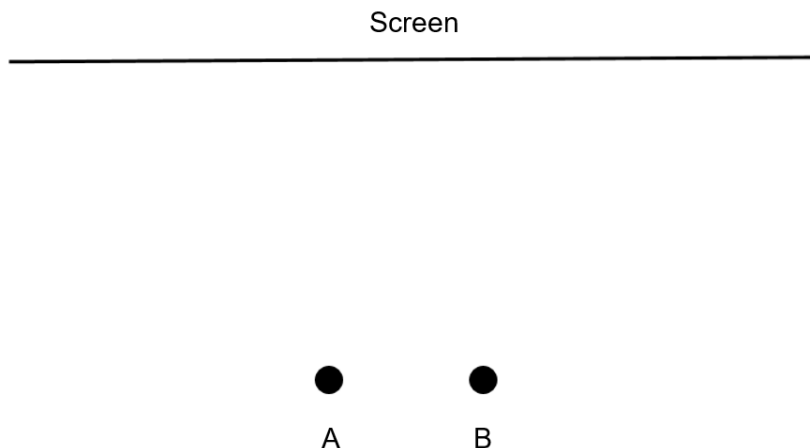


Fig. 5.1

The light from the light sources meet on the screen and a steady interference pattern is formed on the screen.

State two other conditions required for the interference pattern to be observable.

1. Any two of the following:
 - Polarised along the same plane or not polarized [B1]
 - (Roughly) same amplitude/ Intensity [B1]
 - Light source must be coherent [B1]
2.[1]

- (b) When a distant streetlight, which is behaving as a point source of light of wavelength $4.5 \times 10^{-7} \text{ m}$, is viewed through a nylon net curtain, the diffraction pattern of the light projected on a screen is shown in Fig. 5.2. The screen is 3.0 m away from the nylon net curtain.

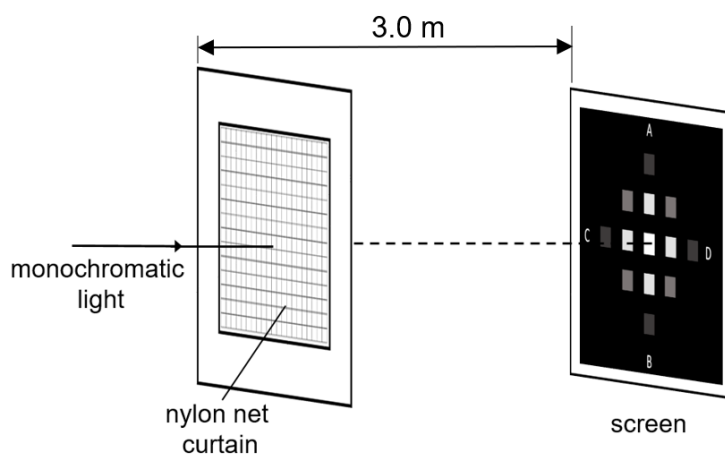


Fig. 5.2

The full-scale diagram of the diffraction pattern is shown in Fig. 5.3.

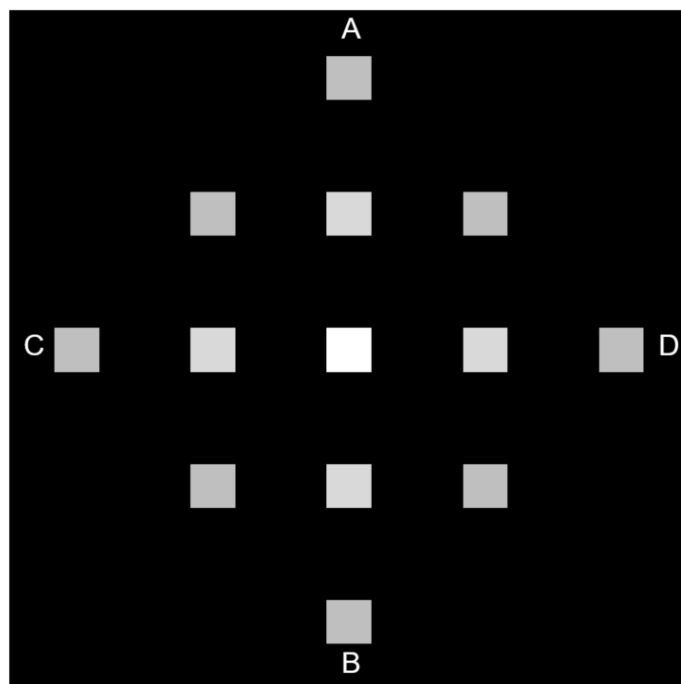


Fig. 5.3

The main feature of this pattern is two lines (AB and CD) of bright images.

- (i) Calculate the angle, in radians, between the orders of the diffracted light.

$$\text{Distance between orders} = 1.8 \text{ cm} \quad [\text{C1}]$$

$$\tan \theta = \frac{0.018}{3.0}$$

$$\theta = 0.0060 \text{ rad} \quad [\text{A1}]$$

$$\text{angle} = \dots\dots\dots \text{rad} \quad [2]$$

- (ii) Using your answer to (b)(i), determine the number of nylon threads per millimetre of the mesh.

$$d \sin \theta = n \lambda$$

$$d \sin (0.0060) = 1 (4.5 \times 10^{-7}) \quad [\text{C1}]$$

$$d = 7.5 \times 10^{-5} \text{ m}$$

$$\text{number per mm} = \frac{1}{7.5 \times 10^{-5}} \times 10^{-3} = 13 \quad [\text{A1}]$$

$$\text{number} = \dots\dots\dots \text{mm}^{-1} \quad [2]$$

- (c) A long horizontal tube, containing fine powder, is closed at one end. A loudspeaker, connected to a signal generator, is positioned at the other end as shown in Fig. 5.4.

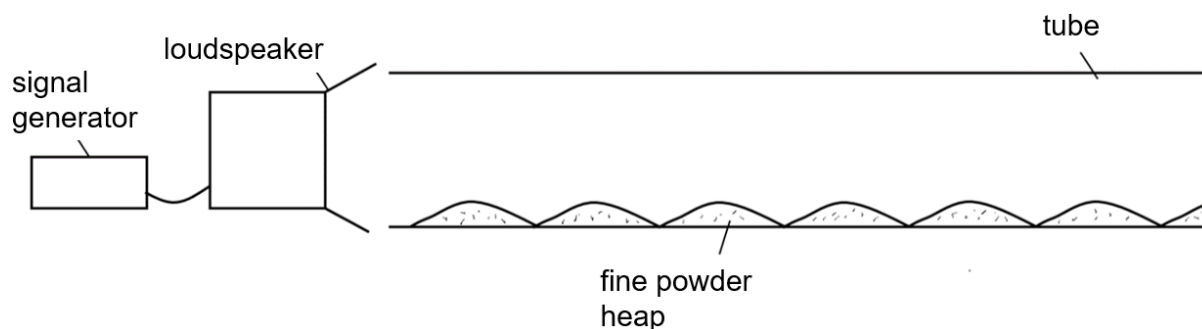


Fig. 5.4

At a particular frequency, a stationary wave is set up inside the tube and the powder forms heaps as shown. The speed of sound is 330 m s^{-1} .

- (i) On Fig. 5.4, mark out 2 points where displacement nodes are and label them as N. [1]
 Any 2 points directly above the heap. [B1]
- (ii) Determine the distance between adjacent heaps if the signal generator is producing a signal with frequency of 3.5 kHz.

$$v = f\lambda$$

$$\lambda = \frac{330}{3.5 \times 10^3} = 0.09429 \text{ m} \quad [\text{M1}]$$

Distance between adjacent heaps is 0.5λ

$$0.5 \times 0.09429 = 0.047 \text{ m} \quad [\text{A1}]$$

spacing = m [2]

[Total: 9]

